

NANOTECHNOLOGY AND APPLICATIONS

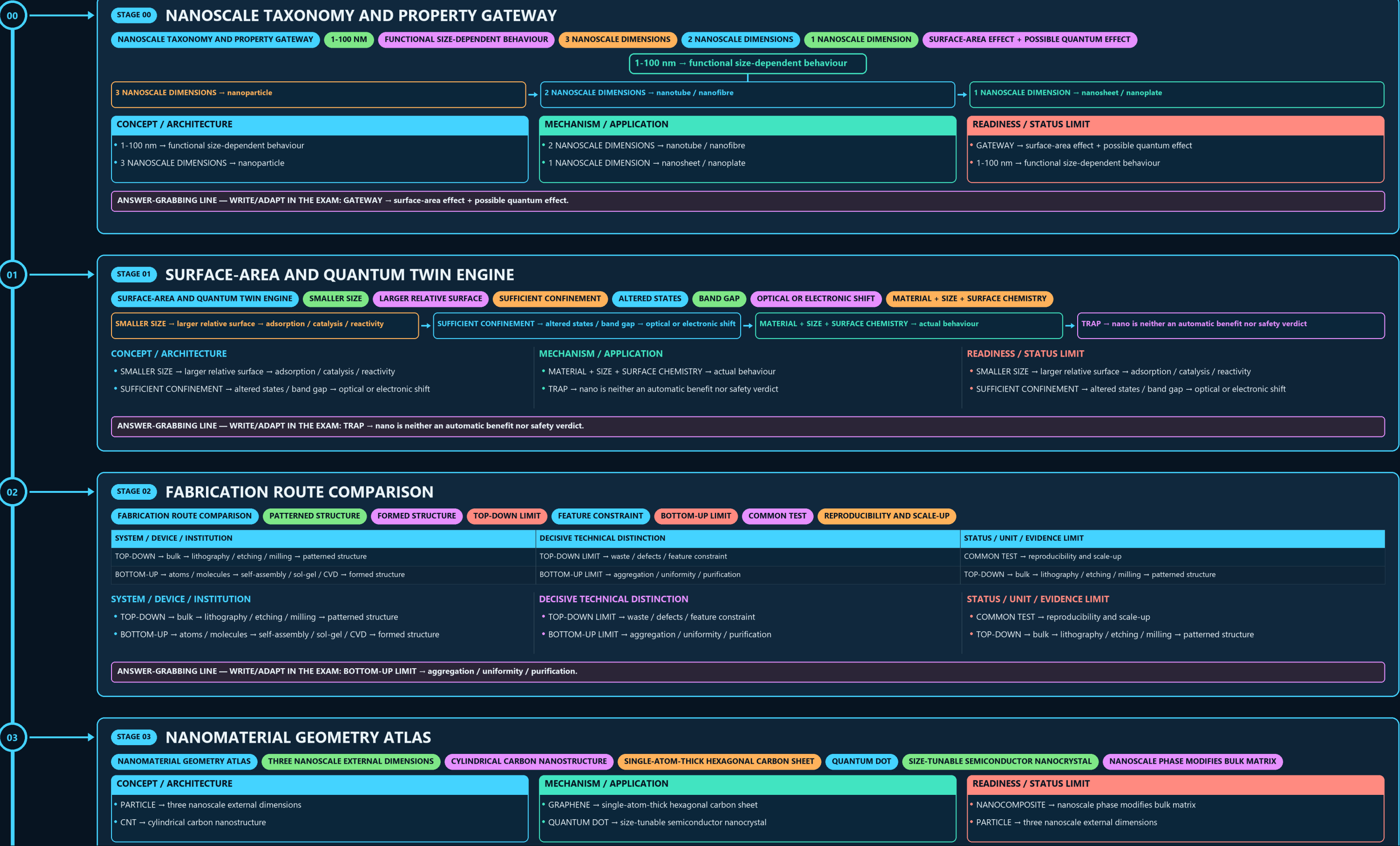
NANOSCALE TAXONOMY AND PROPERTY → SURFACE-AREA AND QUANTUM TWIN → FABRICATION ROUTE COMPARISON → NANOMATERIAL GEOMETRY ATLAS → CHARACTERIZATION DECISION TREE → NANOMEDICINE BARRIER RAIL

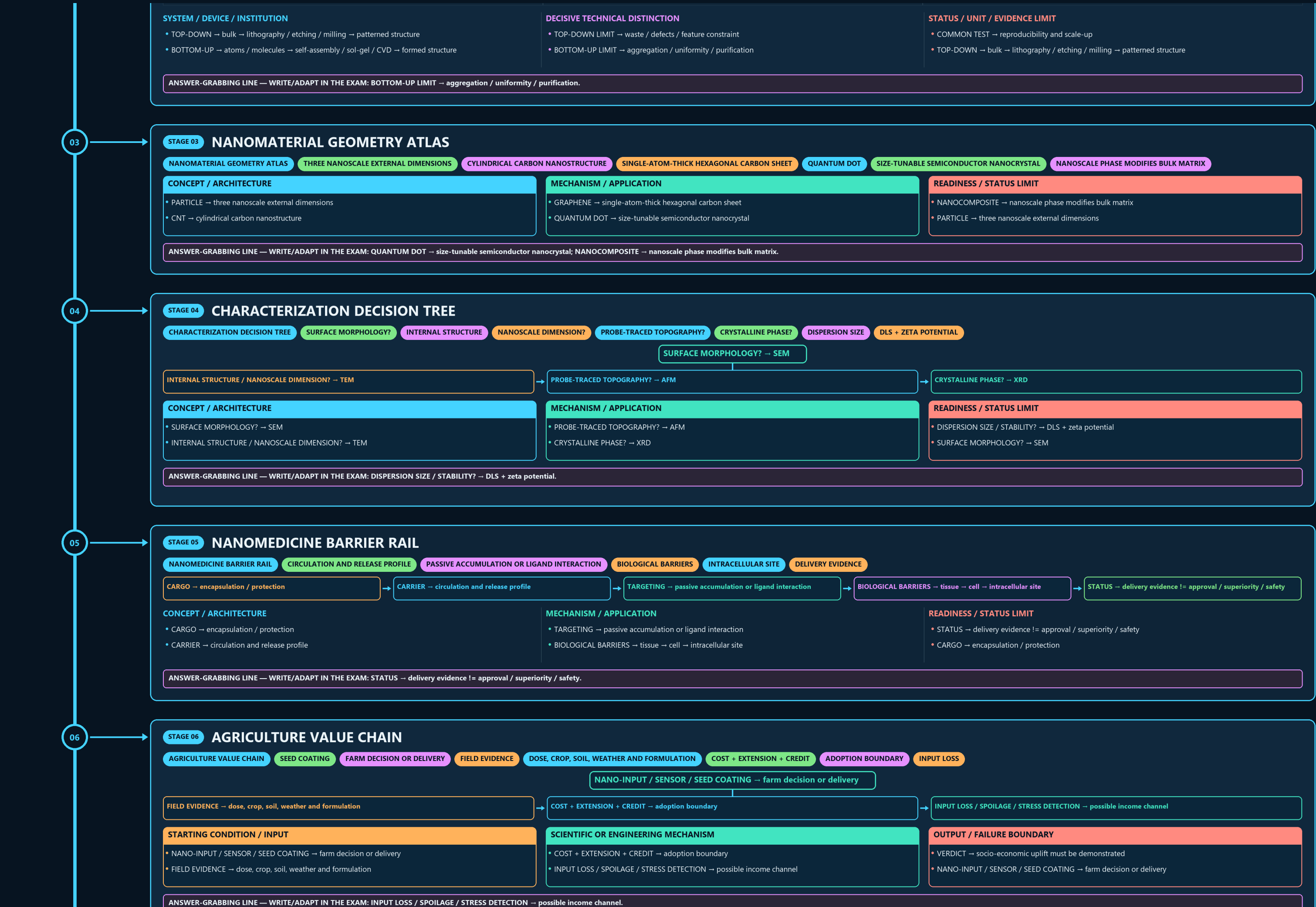
Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

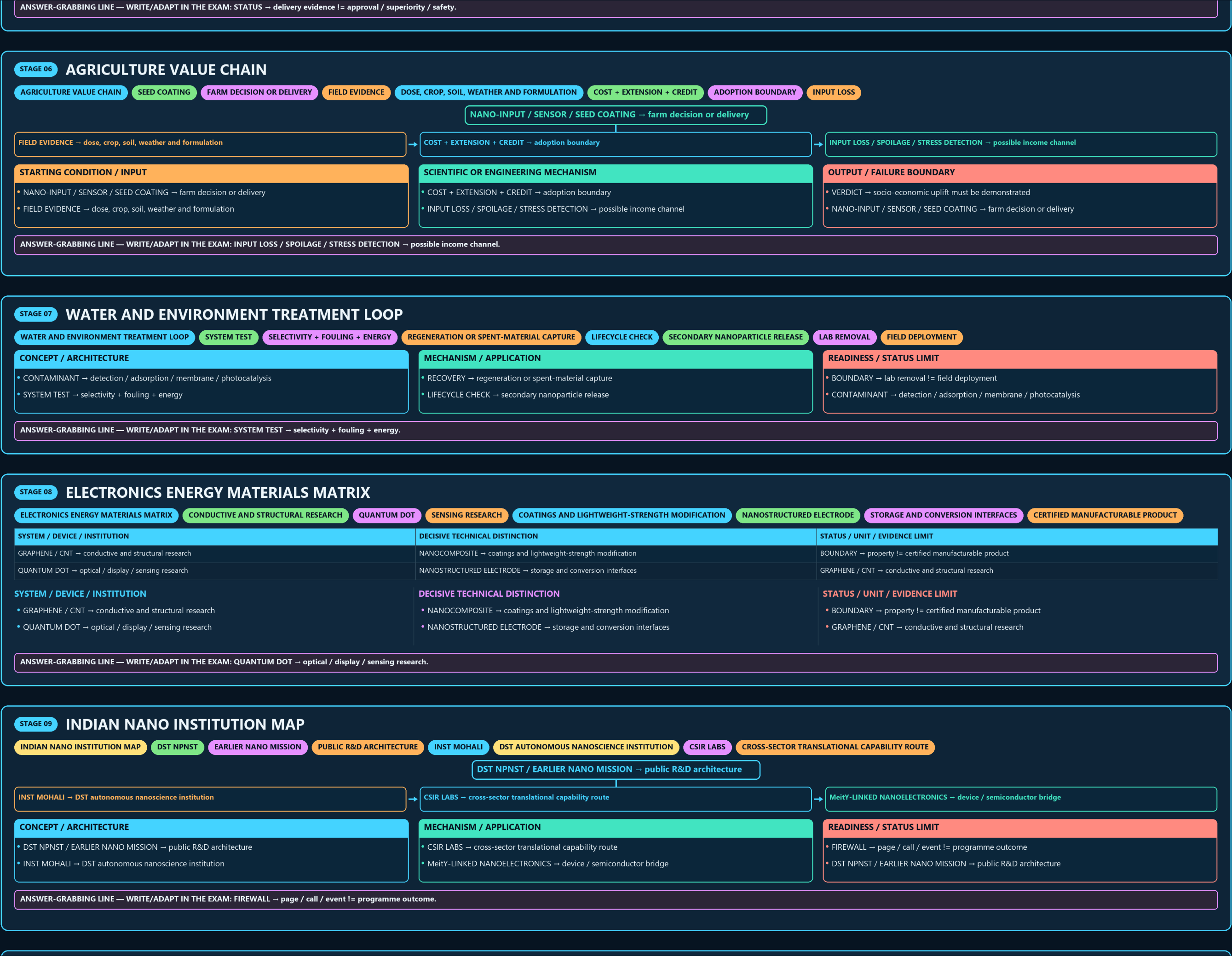
PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles







08

STAGE 08

ELECTRONICS ENERGY MATERIALS MATRIX

ELECTRONICS ENERGY MATERIALS MATRIXCONDUCTIVE AND STRUCTURAL RESEARCHQUANTUM DOTSENSING RESEARCHCOATINGS AND LIGHTWEIGHT-STRENGTH MODIFICATIONNANOSTRUCTURED ELECTRODESTORAGE AND CONVERSION INTERFACESCERTIFIED MANUFACTURABLE PRODUCT

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
GRAPHENE / CNT → conductive and structural research	NANOCOMPOSITE → coatings and lightweight-strength modification	BOUNDARY → property != certified manufacturable product
QUANTUM DOT → optical / display / sensing research	NANOSTRUCTURED ELECTRODE → storage and conversion interfaces	GRAPHENE / CNT → conductive and structural research

SYSTEM / DEVICE / INSTITUTION

- GRAPHENE / CNT → conductive and structural research
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUANTUM DOT → optical / display / sensing research.

09

STAGE 09

INDIAN NANO INSTITUTION MAP

INDIAN NANO INSTITUTION MAPDST NPNSTEARLIER NANO MISSIONPUBLIC R&D ARCHITECTUREINST MOHALIDST AUTONOMOUS NANOSCIENCE INSTITUTIONCSIR LABSCROSS-SECTOR TRANSLATIONAL CAPABILITY ROUTE

INST MOHALI → DST autonomous nanoscience institution

CSIR LABS → cross-sector translational capability route

MeitY-LINKED NANO ELECTRONICS → device / semiconductor bridge

CONCEPT / ARCHITECTURE

- DST NPNST / EARLIER NANO MISSION → public R&D architecture
- INST MOHALI → DST autonomous nanoscience institution

MECHANISM / APPLICATION

- CSIR LABS → cross-sector translational capability route
- MeitY-LINKED NANO ELECTRONICS → device / semiconductor bridge

READINESS / STATUS LIMIT

- FIREWALL → page / call / event != programme outcome
- DST NPNST / EARLIER NANO MISSION → public R&D architecture

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FIREWALL → page / call / event != programme outcome.

